

Course Code: CSE-1012	Course Title: Problem Solving using Python	T P C	3	2	4
Version No.	1.0				
Course Pre-requisites/ Co-requisites	None				
Anti-requisites (if any).	None				
Objectives:	1. To understand why Python is a useful scripting language for developers. 2. To learn how to design and write programs in Python 3. To understand and apply functions with operators, basic data structures, branching and looping constructs. 4. To learn type conversions and String Operations				
Course Outcomes	Course Outcome Statement	PO's / PEO's			
CO1	Develop solutions through algorithms and flowcharts	PO1 PO2 PO3 PO4 PO5			
CO2	Implement simple programs using expressions and statements	PO1 PO2 PO3 PO5			
CO3	Apply the concepts of looping statements and regular expressions to solve complex problems	PO1 PO2 PO3 PO4 PO5 PO6			
CO4	Analyze the concepts of data structures to deploy solutions for real time applications	PO1 PO2 PO3 PO4 PO5 PO6			
CO5	Decide the usage of user defined functions, recursive solutions and modules based on the application domain	PO1 PO2 PO3 PO4 PO5 PO6			
CO6	Create applications based on the file processing and string concepts	PO1 PO2 PO3 PO4 PO5 PO6			
		TOTAL HOURS OF INSTRUCTIONS: 45			
Module No. 1	Introduction to Problem Solving	6 Hours			
Introduction to computers, algorithms, Flow Charts, Problem solving concepts: exchanging the values of two variables, Counting, Summation of a set of numbers, Factorial computation, Sine function computation, Generation of the Fibonacci sequence, Reversing the digits of an integer, Base conversion and character to number conversion.					

Module No. 2	Operators and Branching	7 Hours
Variables, Data types, Arithmetic Operators, Comparison (Relational) Operators, Assignment Operators, Logical Operators, Bitwise Operators, Membership Operators, Identity Operators, Branching Statements-if; ifelse, nested if; nested ifelse;		
Module No. 3	Loops and Regular Expressions	8 Hours
Creating Loops with while and for, Different versions of Loops, Nested Loops, Loop Control Statements, Loop Modification with break, continue and pass, Regular Expressions.		
Module No. 4	Basic Data Structures	8 Hours
Creating, Accessing and Manipulating Lists, Sets, Tuples and Dictionaries, Understanding the differences among them, Applications of the Data Structures. Using Branching and Control loops with Data structures, Matrix Operations using Numpy.		
Module No. 5	Functions	8 Hours
Pre-defined functions, User defined functions, formal and actual parameters, return statement, Using Branching, Looping and Data structures in Functions, Recursion, Internal workflow of Recursion, Modules.		
Module No. 6	Working with Numbers, Strings and Files	8 Hours
Introduction to Different Numeric Types, Type Conversion, Mathematical Functions, Random Numbers. Creating and Accessing Strings, Operations on Strings, Indexing, Slicing, String Manipulations, Pre-defined functions on Strings.		
File I/O-Opening and Closing files, Different modes, File attributes, Read, Write Operations, File Positions. Renaming and Deleting Files, various directory handling functions.		
Text Books 1. Kenneth Lambert, “Fundamentals of Python: First Programs”, Cengage Learning, 2019 2. Martin C Brown, “The Complete Reference with Python”, McGrawHill, 2018.		
References 1. John Zelle, Python Programming: An introduction to Computer Science, Franklin Associates, Third Edition, 2016. 2. Mark Lutz, “Learning Python”, Fifth edition, O’Reilly, 2013. 3. R.G. Dromey, “How to Solve it by Computer”, Pearson, 1982.		
List of Laboratory Experiments		
Week 1:		

1. Write a Python program to display the current date and time.
2. Write a Python program to get the Python version you are using
3. Write a Python program that accepts an integer (n) and computes the value of $n+nn+nnn$
4. Write a Python program to read and print various types of variables.
5. Write a Python program to print the calendar of a given month and year.

Week 2:

1. Python Program to Find the Square Root
2. Python Program to Calculate the Area and Perimeter of Triangle and Circle.
3. Python Program to Solve Quadratic Equation
4. Python Program to Swap Two Variables
5. Python Program to Convert Kilometres to Miles
6. Python Program to Convert Celsius To Fahrenheit

Week 3:

1. Python program to find whether the given number is Even or Odd
2. Write a Python program to get the difference between a given number and 17, if the number is greater than 17 return double the absolute difference
3. Write a Python program to test whether a number is within 100 of 1000 or 2000.
4. Write a Python program to calculate the sum of three given numbers, if the values are equal then return three times of their sum
5. Python Program to Find the Factorial of a Number
6. Python Program to print maximum of 3 numbers
7. Write a python program to find whether a given year is leap or not.

Week 4:

1. Write a program which will find all such numbers which are divisible by 7 but are not a multiple of 5, between 2000 and 3200 (both included).
The numbers obtained should be printed in a comma-separated sequence on a single line.
Consider use range(begin, end) method
2. Write a python program to check whether a number is divisible by 5 and 11 or not.
3. Write a python program to check whether a character is alphabet or not.
4. Write a python program to input any character and check whether it is alphabet, digit or special character.
5. Write a python program to check whether a character is uppercase or lowercase alphabet.
6. Write a python program to input week number and print week day.
7. Write a python program to count total number of notes in given amount

Week 5:

1. Write a Python program to print all natural numbers from 1 to n. - using while loop
2. Write a Python program to find sum of all odd numbers between 1 to n.
3. Write a Python program to count number of digits in a number.
4. Write a Python program to find first and last digit of a number.
5. Write a Python program to calculate sum of digits of a number.
6. Write a Python program to enter a number and print its reverse.

Week 6:

1. Write a Python program to check whether a number is palindrome or not.
2. Write a Python program to find frequency of each digit in a given integer.
3. Write a Python program to print all ASCII character with their values.
4. Write a Python program to find all factors of a number.
5. Write a Python program to calculate factorial of a number.
6. Write a Python program to print all Prime numbers between 1 to n.
7. Write a Python program to check whether a number is Armstrong number or Strong or Prime Number or Perfect number or magic number or not
8. Write a Python program to print Fibonacci series up to n terms.

Week 7:

1. Write a Python Program to Find the Largest Number in a List
2. Write a Python Program to Find the Second Largest Number in a List
3. Write a Python Program to Put Even and Odd elements in a List into Two Different Lists
4. Write a Python Program to Merge Two Lists and Sort it
5. Write a Python Program to Sort the List According to the Second Element in Sublist
6. Write a Python Program to Find the Second Largest Number in a List Using Bubble Sort
7. Write a Python Program to Sort a List According to the Length of the Elements
8. Write a Python Program to Find the Union of two Lists
9. Write a Python Program to Find the Intersection of Two Lists
10. Python Program to print all odd indexed elements of a list

Week 8:

1. Write a Python program to get the 4th element and 4th element from last of a tuple
2. Write a Python program to find the repeated items of a tuple.
3. Write a Python program to check whether an element exists within a tuple
4. Write a Python program to unzip a list of tuples into individual lists.
5. Write a Python program to replace last value of tuples in a list.
Sample list: [(10, 20, 40), (40, 50, 60), (70, 80, 90)]
Expected Output: [(10, 20, 100), (40, 50, 100), (70, 80, 100)]
6. Write a Python program to remove an empty tuple(s) from a list of tuples.

Sample data: [((), ()), ('', ('a', 'b')), ('a', 'b', 'c'), ('d')]

Expected output: [('', ('a', 'b')), ('a', 'b', 'c'), ('d')]

7. Write a Python program to convert a list of tuples into a dictionary.

8. Write a Python program to find the highest 3 values of corresponding keys in a dictionary.

Week 9:

1. Write a Python function that prints out the first n rows of Pascal's triangle.

2. Write a Python function to create and print a list where the values are square of numbers between 1 and 30 (both included).

3. Write a Python program to detect the number of local variables declared in a function

4. Write a Python program that invoke a given function after specific milliseconds.

5. Write a Python program to get the sum of a non-negative integer

Week 10:

1. Write a Python program to calculate the harmonic sum of n-1

2. Write a Python program to calculate the sum of the positive integers of $n+(n-2)+(n-4)\dots$ (until $n-x \leq 0$)

3. Write a Python program to find the greatest common divisor (gcd) of two integers using Recursion

4. Implement any sorting algorithm using Recursion.

Week 11:

1. Write a Python Program to Replace all Occurrences of 'a' with 'b' in a String. If 'a' is not present then print appropriate message.

2. Write a Python Program to Remove the nth Index Character from a Non-Empty String

3. Write a Python Program to Detect if Two Strings are Anagrams

4. Write a Python Program to Form a New String where the First Character and the Last Character have been Exchanged.

Week 12:

1. Write a Python program to get the last part of a string before a specified character

2. Write a Python program to count the occurrences of each word in a given sentence.

3. Write a Python function to insert a string in the middle of a string.

4. Write a Python function to get a string made of its first three characters of a specified

string. If the length of the string is less than 3 then return the original string.

5. Write a Python program to add a prefix text to all of the lines in a string

6. Write a Python program to convert a given string into a list of words.

A few Programming exercises for Competitive Coding (optional) :

1. Remove Duplicate Char from String
2. Hailstone Sequence
3. Secure Conversation by Encryption and Decryption
4. Special Elements in Matrix
5. Next Greater No with the Same set of Digits
6. Smallest Subarray with Sum Greater than Given Number
7. Group Anagrams
8. Find Duplicates in Array in $O(n)$
9. Find Two Unique Numbers from Array in $O(n)$
10. Number Patterns & Finding Smallest Number
11. Minimum Distance for Truck to Deliver Order [Amazon]
12. Generate Balanced Parentheses

Mode of Evaluation	Cumulative Lab Assessment	20%
	Continuous Assessment Test-1	20%
	Continuous Assessment Test-2	20%
	Final Assessment Test	20%
	Practical Assessment	20%
Recommended by the Board of Studies on		
Date of Approval by the Academic Council		